Yazhou Tang

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Education

Zhejiang University 2025/09 — 2028/06 (expected)
Master of Engineering in Computer Techonology Hangzhou, Zhejiang, China
- Member of ZJU Formal Verification Group in ZJU Advanced Computing and Emerging Service (ACES) Lab
- Supervised by Mingshuai Chen and co-advised by Shenghao Yuan

Shanghai Jiao Tong University 2019/09 — 2023/06
Bachelor of Engineering in Computer Science and Technology Shanghai, China
- Honors: SJTU Outstanding Bachelor's Graduate Award (Top 15%)
- Selected Courses: Programming Languages, Program Analysis & Verification, Operating Systems, Compiler Principles

Publications

* Equal contribution. ✉ Corresponding author.

[1] Shenghao Yuan*, **Yazhou Tang***, Tianci Cao, Frédéric Besson, Jean-Pierre Talpin, and Mingshuai Chen✉. 2026. Formalizing the Linux eBPF Core ISA: A Mechanized Operational Semantics and Its Real-World Applications. *Proc. ACM Program. Lang.* 10, OOPSLA2 (October 2026). https://doi.org/10.1145/3839449

[2] **Yazhou Tang**, Shenghao Yuan✉, Jean-Pierre Talpin, and Mingshuai Chen✉. 2026. Formal Specification of Linux eBPF Instruction Set Architecture in Sail. In *Proceedings of the 4th Workshop on EBPF and Kernel Extensions (eBPF'26)*, 2026. Association for Computing Machinery, Prague, Czech Republic, 26–32. https://doi.org/10.1145/3837779.3838167

Research Experience

Formalizing the Semantics of the Linux eBPF ISA [OOPSLA'26 Paper & Slides] 2025/06 — Now
Advisor: Shenghao Yuan | Collaboration with Inria Rennes Hangzhou, Zhejiang, China
- Formalized all 153 sequential in-kernel Linux eBPF instructions as executable small-step semantics in Rocq Prover using CompCert's memory model. Built a validation pipeline comparing per-step register states against kernel execution traces from 1,700+ Linux selftests.
- Proved the soundness of all 32-bit and 64-bit tristate number (tnum) abstract operations using the formal semantics. Identified and fixed bugs in the Linux kernel's eBPF interpreter and verifier during formalization. Upstreamed these fixes and eBPF verifier precision improvements for byte-swap and division operations.
- Follow-up: Developed an eBPF ISA model in Sail, generating Rocq definitions and a human-readable specification of instruction encodings and operational semantics from a single source. Currently working on conformance testing and semantic equivalence proofs. [eBPF'26 Paper & Slides]

VST-IDE: An Interactive Tool for Program Verification [Bachelor Thesis & Slides] 2021/10 — 2023/06
SJTU Undergraduate Research & Thesis Project | Advisor: Qinxiang Cao Shanghai, China
- Integrated Verified Software Toolchain (VST) into a C development environment to automate memory-safety verification using preconditions, postconditions, and assertions written as code annotations.
- Developed a separation-logic SMT solver supporting custom predicates for entailment checking between symbolic execution states and user assertions, using the frame rule and predicate splitting.
- This work served as an early exploration for the publicly released **QCP Project**. [Official Website]

Verification of Red-Black Tree's Source Code Safety [Project Report & Slides] 2021/01 — 2021/09
SJTU Undergraduate Research | Advisor: Qinxiang Cao Shanghai, China
- Verified insertion, deletion, and lookup in a C red-black tree using Rocq and VST against an abstract map specification.
- Modeled parent pointers with half-tree and partial-tree representations to resolve cyclic dependencies, and separated nodes from stored values to verify lookups returning value addresses.

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Work Experience

Bilibili Inc. 2023/07 — 2024/10

Multimedia Software Engineer (Playback Quality Group) Shanghai, China

- Contributed to core module development in bilibili/ijkplayer and customized FFmpeg components, optimizing mobile VOD/live playback through performance tuning and A/B testing.
- **AV1 Hardware Decoding**: Developed mobile AV1 decoding using Android MediaCodec and iOS VideoToolbox, and collaborated with MediaTek/Xiaomi engineers to resolve issues across 10+ device models. Enabled AV1 live streaming for popular live rooms, reducing bandwidth consumption by 30% compared with H.265.
- **Audio Normalization**: Built EBU R128-based loudness control with FFmpeg filters, reducing initial audio sampling from 3s to 400ms and first-frame processing latency by 85%. Deployed to all mobile VOD users.

Competitions

2nd Prize, CCF ChinaSoft Theorem Proving Competition [Official Website] 2025/11

2nd Prize, SJTU Android App Development Competition [Code] 2021/12

Excellence Award, World Robot Contest Finals (WRCF) VEX-U Track [News] 2020/12

Honors & Awards

Five-A Graduate Student Award, Zhejiang University 2026/10

Outstanding Graduate Student Award, Zhejiang University 2026/10

Outstanding Bachelor's Graduate Award, Shanghai Jiao Tong University 2023/06

Academic Improvement Scholarship, Shanghai Jiao Tong University 2022/12

Outstanding Student Scholarship (Category C), Shanghai Jiao Tong University 2020/12

Triple-A Student Award, Shanghai Jiao Tong University 2020/11

Skills

- **Programming Languages**: Rocq Prover (Coq), OCaml, C/C++, Rust, Python
- **Formal Methods**: Programming Language Theory (PLT), Formal Verification, Interactive Theorem Proving
- **System Engineering**: Linux Kernel Development (See my contribution in upstream), eBPF

Academic Services

- **Artifact Evaluation Committee (AEC) Member**: OOPSLA 2026.
- **Sub-Reviewer**: OOPSLA 2026, CAV 2026, TACAS 2027.
- **Student Volunteer**: CCF Academic Forum on Network and Systems Security (Jan. 2026@ZJU), CCF Youth Academic Forum on Formal Methods (May. 2026@ZJU)

Miscellaneous

- **Languages**: Chinese (Native), English (Working Language)
- **Sports**: Football, Running, Table Tennis, Badminton

Last Updated on October 29, 2026